MULTIMODAL • LONGITUDINAL • AI-ASSISTED

HEALTH INNOVATION PROTOTYPE

NEXT-GEN HEALTH INTELLIGENCE

NUVYRA

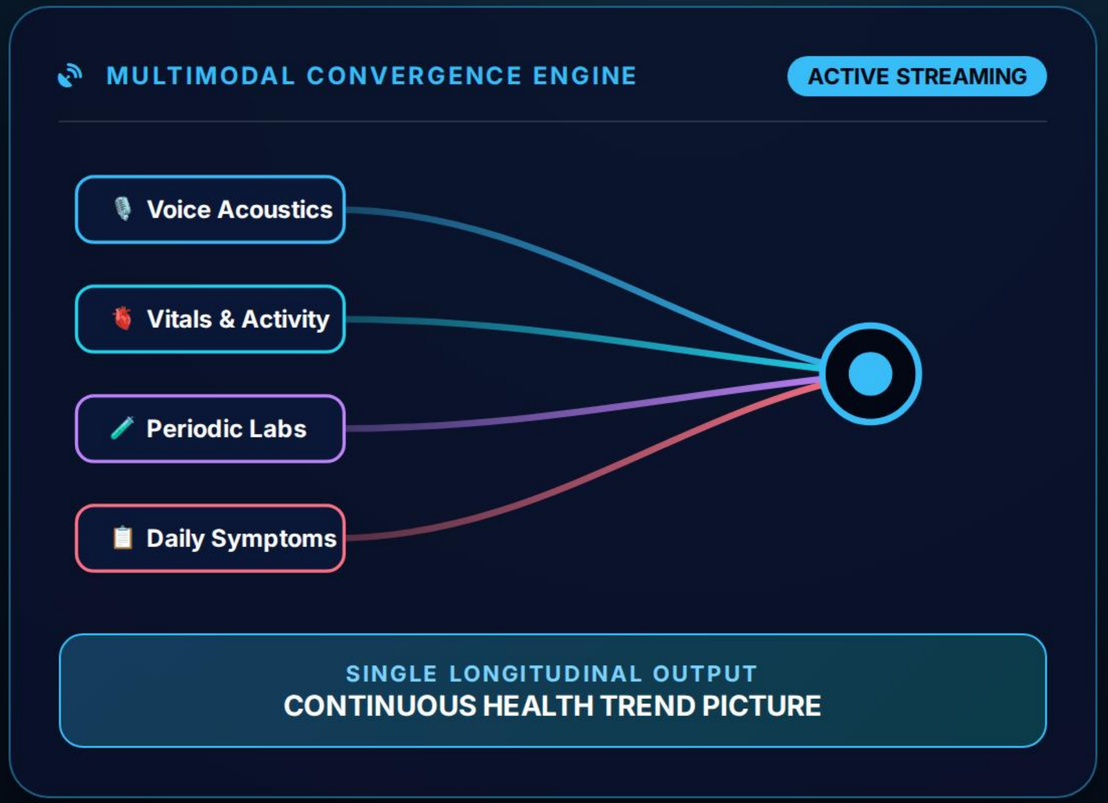
AI-Powered Longitudinal Biomarker Intelligence

> *"Turning everyday health signals into a clearer picture of change."*

A digital health platform bridging critical information gaps between intermittent clinical appointments through multi-signal temporal convergence.

PS Lead Developer: **Prapti Sharma**

Live Web Prototype



HUMAN-CENTERED PROBLEM STORY

Healthcare sees moments. Health happens continuously.

A typical 6-month patient journey captures less than 0.01% of daily health signals.

Maya's 180-Day Health Journey (Hypothetical Patient Case)

180 DAYS • 2 CLINIC VISITS

MON	TUE	WED	THU	FRI	SAT	SUN
🎙️	💊	🏥	🫀	📋	🌙	🧪
Voice Fatigue	Med Missed	15-Min Visit	BP Elevation	Tremor Flare	Sleep Loss	Lab Result
Unrecorded	Unrecorded	Recorded	Unrecorded	Unrecorded	Unrecorded	Isolated

"A patient may feel subtly different every day — but most clinical records capture only selected, isolated moments. Crucial early-warning trajectory changes pass unnoticed in the dark."

DATA ARCHITECTURE CHALLENGE

The problem isn't a lack of data. It's disconnected data.

Bridging fragmented healthcare silos into one continuous longitudinal stream.

⚠ TRADITIONAL FRAGMENTED VIEW

SILOED & ISOLATED

- 📄 Isolated Lab Reports
- 💊 Unlinked Rx Regimen
- 🫀 Episodic Vitals
- 📋 Subjective Notes

Clinicians must manually cross-reference 5+ disconnected portals while under 15-minute time constraints.

NUVYRA INTEGRATED SOLUTION

UNIFIED CONVERGENCE

ONE LONGITUDINAL HEALTH PICTURE

Continuous Temporal Correlation

NUVYRA is designed to organize multiple everyday and clinical signals into a single, structured, chronological timeline for instant clinical interpretation.

PRODUCT DEFINITION

NUVYRA connects the signals between visits.

An end-to-end multimodal pipeline designed to structure longitudinal observations.

"NUVYRA is an AI-assisted multimodal health monitoring platform designed to organize longitudinal patient information and identify meaningful trajectory changes across multiple everyday health signals."



Designed to complement — not replace — clinical assessment.

MULTIMODAL SIGNAL MATRIX

One signal can be noisy. Patterns across signals inform.

Combining seven key health streams to isolate genuine health trajectories.

🎙️ VOICE ACOUSTICS

Speech tempo, acoustic shimmer/jitter, vocal pause duration.

👁️ FACIAL DYNAMICS

Blink rate, micro-expression symmetry, eye aperture changes.

🫀 DAILY VITALS

Heart rate trends, SpO2, blood pressure, weight changes.

📋 SYMPTOMS

Patient-reported severity, flare-up timing, body map localization.

🧪 CLINICAL LABS

Periodic metabolic panels, inflammatory markers, HbA1c trajectory.

💊 MEDICATION

Adherence tracking, dose timing, prescription adjustments.

🍎 LIFESTYLE & ENVIRONMENT

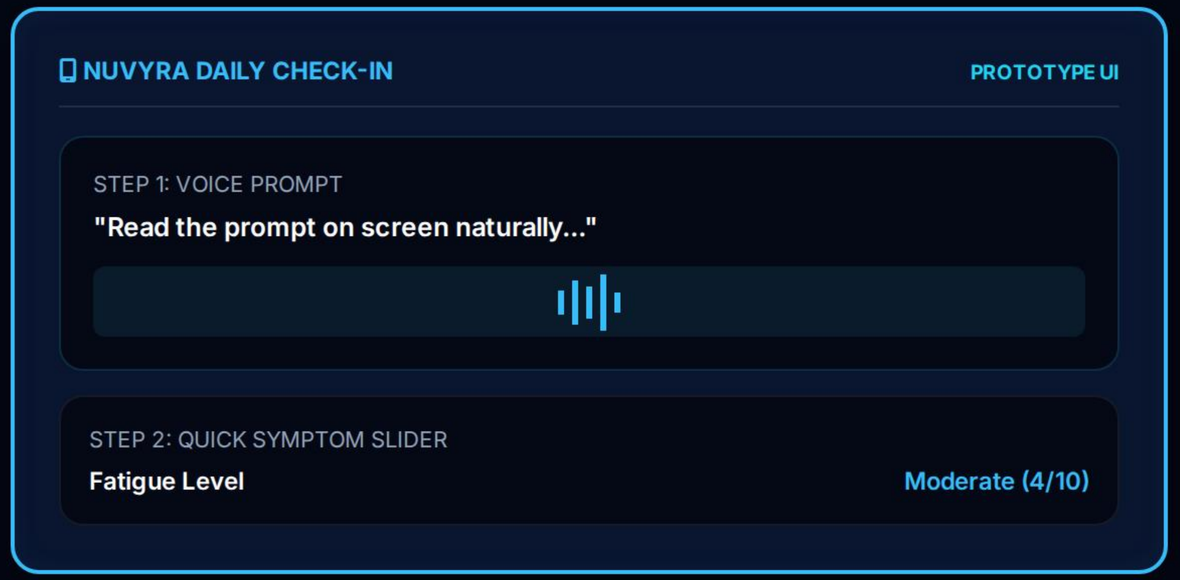
Sleep quality, daily physical activity, nutrition context.

Core Principle: NUVYRA evaluates multi-dimensional signal patterns rather than relying on any single isolated measurement.

PATIENT-FACING EXPERIENCE

Make monitoring part of everyday life.

A simple, friction-free 60-second check-in routine designed for high long-term adherence.



1. **Open NUVYRA App**
   Daily automated notification prompts user check-in.
2. **Record 10-Second Acoustic Prompt**
   Captures research-level speech acoustics & facial movement markers.
3. **Log Daily Symptoms & Vitals**
   Fast tap UI for symptoms, BP, heart rate, and medication status.

RESEARCH FEATURE NOTE: Voice & facial signal extraction are presented as research prototype features designed for longitudinal study.

TECHNICAL FEATURE EXTRACTION

From raw signals to meaningful features.

How NUVYRA processes heterogeneous input streams into normalized trajectory metrics.

1. AUDIO PROCESSING

Acoustic Features

- Pitch jitter & shimmer
- Fundamental frequency (F0)
- Pause-to-speech ratio

2. VIDEO DYNAMICS

Facial Movements

- Blink duration & frequency
- Eyebrow elevation symmetry
- Lip corner depression rate

3. CLINICAL ANALYTICS

Biomarker Slopes

- eGFR delta over time
- HbA1c moving averages
- BP variance index

4. BEHAVIORAL LOG

Symptom Clustering

- Severity progression curves
- Medication timing gaps
- Activity drop correlates

LONGITUDINAL TEMPORAL ALIGNMENT

NUVYRA tracks how features evolve over time relative to a patient's personal baseline.

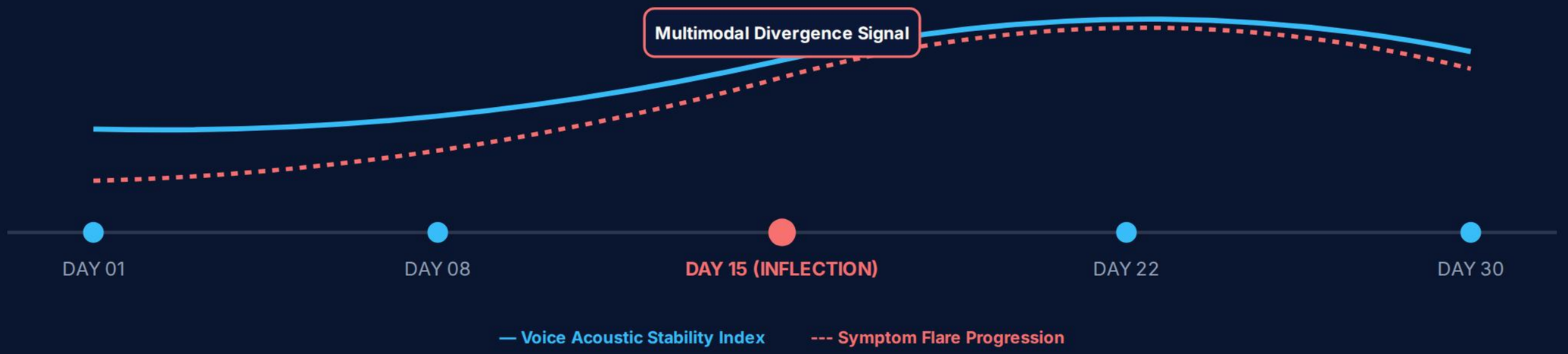
LONGITUDINAL VISUALIZATION

The question isn't 'What is the value today?'

It's 'How is it changing?'

30-DAY MULTIMODAL TRAJECTORY GRAPHIC

ILLUSTRATIVE PROTOTYPE DATA



CLINICIAN-FACING DASHBOARD

Less fragmentation. More context.

Consolidating longitudinal evidence into an actionable 30-second clinical view.

MS

Maya S. (ID: #49201)
42 y/o Female • Longitudinal Monitoring Active

HEALTH STABILITY: STABLE WITH DRIFT

30-DAY VITALS SUMMARY
128/82 mmHg
Mean HR: 72 bpm (Normal)

MEDICATION ADHERENCE
94% Compliance
2 missed doses logged

VOICE ACOUSTIC DRIFT
+14% Pause Ratio
Slight deviation from baseline

Designed to give clinicians immediate temporal context before stepping into the examination room.

EXPLAINABLE SIGNAL INTELLIGENCE

NUVYRA turns observations into a question.

"What changed?"

CHANGE SIGNAL DETECTED

DRIFT ▲

Multi-signal trajectory variation observed between Day 12 and Day 18.

Underlying Evidence Drill-Down:

- **Voice Trend:** Acoustic pause duration increased by 18%
- **Symptoms:** Daily fatigue rating shifted from 2/10 to 6/10
- **Vitals:** Resting heart rate elevated by 8 bpm

IMPORTANT DISCLAIMER: An experimental prototype-level health stability/change signal designed to summarize longitudinal patterns for professional clinical review.

TEMPORAL CONTEXT INTEGRATION

Context matters.

Merging clinical records, daily medication routines, and lifestyle factors.

🏥 CLINICAL DATA

- Laboratory test records
- Clinical vital measurements
- Physician consultation notes

💊 MEDICATION

- Prescribed dosing regimen
- Daily adherence timestamps
- Side-effect self-reports

🍎 LIFESTYLE & DIET

- Daily nutrition context
- Physical movement logs
- Subjective sleep ratings

UNIFIED TEMPORAL FRAMEWORK

NUVYRA places all data streams into the exact same temporal context for accurate correlation.

SECURITY & PRIVACY ARCHITECTURE

Health intelligence requires trust first.

Multi-layered security protocols built directly into the system architecture.

🔐

JWT Auth

Stateless session tokens & secure authorization

🛡️

Bcrypt Hashing

Salted password hashing layer

👥

RBAC Access

Strict Patient vs Clinician role boundaries

📦

Data Isolation

Row-level tenant data separation

🔒 Security and privacy are treated as core foundational architecture, not an afterthought.

FULL-STACK ENGINEERING

Built as a real prototype — not just a concept.

Modern production-ready open web technologies powering the NUVYRA platform.

FRONTEND

React + TS

Vite, Axios, Tailwind CSS, React Router

BACKEND

FastAPI

Python, SQLAlchemy, Pydantic, JWT

DATABASE

PostgreSQL

Relational longitudinal data schema

DEPLOYMENT

Render Cloud

Automated continuous deployment

DATA FLOW PIPELINE

User App ➔ FastAPI REST Endpoints ➔ PostgreSQL DB ➔ Multimodal Trajectory Processing

WORKING PROTOTYPE DEMONSTRATION

This isn't just an idea. It's a working prototype.

Key functional modules built and deployed in the live NUVYRA web application.

1. Interactive Dashboard
Live health stability cards, vitals feed, and alert feeds.

2. Daily Check-In Module
Voice recording UI, symptom ratings, and daily logs.

3. Longitudinal Timeline
30-day chronological event & signal correlation graphs.

4. Clinical Labs Tracker
Structured lab uploads and longitudinal marker trend lines.

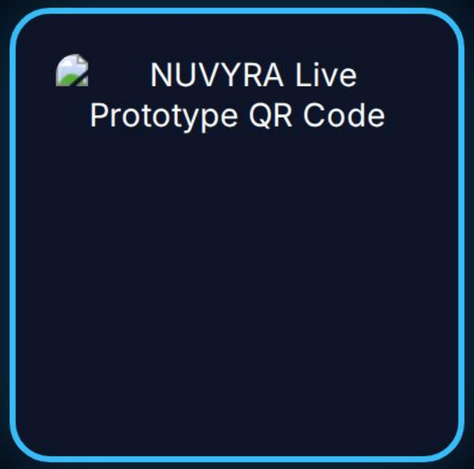
5. Medication & Lifestyle
Regimen compliance tracking and daily nutrition logs.

6. Patient & Role Switcher
Multi-role toggle between Patient and Clinician view modes.

LIVE DEMONSTRATION

See NUVYRA in action.

Scan the QR code or click the live URL to explore the deployed working prototype.



https://digital-biomarker-monitor.onrender.com

Web Prototype • Deployed Live on Render Cloud

VALUE PROPOSITION

From fragmented observations to understanding.

Transforming healthcare data from disconnected snapshots into a continuous story.

BEFORE NUVYRA

- ❌ Fragmented snapshot visits
- ❌ Disconnected portal silos
- ❌ Reactive intervention after escalation
- ❌ High cognitive burden on clinicians

WITH NUVYRA

- ✅ Continuous longitudinal trajectory
- ✅ Unified multimodal convergence
- ✅ Context-aware early drift signals
- ✅ Streamlined 30-second clinical summary

"NUVYRA is designed to help transform health data from isolated observations into a connected, human story."

RESPONSIBLE INNOVATION

Building responsibly means knowing limits.

Clear boundaries regarding scientific validation, clinical scope, and AI governance.

1. Research Prototype
NUVYRA is an engineering prototype, not a certified medical device.

2. Clinical Validation Needed
Signal algorithms require formal clinical trials before diagnostic deployment.

3. Environmental Variance
Acoustic and video signals can be affected by ambient noise and lighting.

4. Clinician Sovereignty
All health decisions must remain under the direction of qualified medical professionals.

Next Development Step: Validating signal models against real longitudinal clinical datasets.

FUTURE ROADMAP

What could NUVYRA become?

A strategic roadmap for scaling from student prototype to clinical research platform.

STAGE 1 • NOW

Working Prototype

Deployed full-stack application with baseline feature extraction.

STAGE 2 • NEXT

Feasibility Pilots

User experience studies with patient cohorts and clinical advisors.

STAGE 3 • THEN

Clinical Validation

Formal correlation studies against expert-defined clinical outcomes.

STAGE 4 • FUTURE

Scalable Platform

Integration with wearable biosensors & EHR interoperability APIs.

THE NUVYRA VISION

"Health doesn't happen in snapshots."

NUVYRA is built to see the story between them.

NUVYRA
HEALTH INTELLIGENCE
Multimodal • Longitudinal • Human-Centered

Lead Developer: **Prapti Sharma** | Live Demo: digital-biomarker-monitor.onrender.com